Akshara Sarma Chennubhatla

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.31
RV PU College	Class XII (KSEAB)	2024	95.83%
National Public School, Kengeri	Class X (CBSE)	2022	97.60%

SCHOLASTIC ACHIEVEMENTS

- Received **A+** grade(outstanding performance) in Probability (EE1080), Random Processes (EE2080) and Introduction to Statistics (MA2140) courses taken in college.

SKILLS

- **Programming Languages:** Python, C, RISC-V Assembly
- **Hardware Description Language (HDL):** Verilog
- **Tools:** LTSpice, Arduino IDE, MATLAB, Git/GitHub, Latex, Linux, Altium Designer, Intel Quartus Lite
- **Frameworks:** Numpy, Matplotlib, Seaborn, Pandas, Scikit-learn, OpenCV, ROS2, Nav2, Slam ToolBox

RELEVANT COURSES

- **Mathematics:** Math for EE, Matrix Theory, Probability, Random Processes, Numerical Analysis, Intro to Statistics
- **Electrical Engineering:** Digital Systems, Signal Processing
- **Computer Science & AI:** Intro to Programming, Data Structures and Applications, Machine Learning
- **Labs:** Digital Systems Lab, DSP Lab

PROJECTS

- **OpenBPU: A Probabilistic RISC-V Processor for Bayesian Computing |** Github**:**
- Designed and implemented a **5-stage pipelined RV32IM RISC-V processor** in Verilog, evolving from a single-cycle RV32I core with hazard handling, forwarding, pipeline control, **2-bit dynamic branch prediction** and hardware integer multiplication/division.
- Extended the ISA with custom instructions for **Q8.24 fixed-point arithmetic, random sampling, CORDIC-based functions** (log, exp, sqrt, sin, cos, atan2), and **Bayesian primitives** (log.sum.exp, normal.logpdf).
- Developed dedicated hardware modules including a **Ring Oscillator TRNG, Box-Muller Gaussian generator, CORDIC engine**, fixed-point unit, and custom instruction decode/execute pipeline for probabilistic computing.
- Currently integrating an **MMIO-based parallel Blelloch prefix-sum tree** to accelerate Sequential Monte Carlo (SMC) resampling, with deployment targeted on a **PYNQ-Z2 FPGA.**

- **Inter-IIT Tech Meet 14.0 - Autonomous Warehouse Rover |** Github**:**
- Designed and implemented the **ROS2** software architecture on **Raspberry Pi**, including kinematics, wheel odometry for a 4-wheel mecanum-drive and z-axis mechanism nodes.
- Interfaced with **Nav2** and **SLAM Toolbox** stack for autonomous navigation and developed an **OpenCV-based keepout filter** enforcing no-go areas during navigation.
- Developed **ESP32** hardware interfaces, joystick teleoperation, ROS2 launch files, and validated the navigation stack in **Gazebo**.

- **e-Yantra Robotics Competition (eYRC) | Krishi Cobot Theme |** Github**:**
- Developed the **ROS2** software stack for autonomous navigation, implementing **LiDAR**-based localization, wheel odometry, **PID control**, and waypoint navigation for the **eBot**.
- Built perception using **OpenCV, ArUco markers**, and LiDAR for fertilizer pickup, geometric shape detection, and HSV-based fruit quality classification
- Integrated and tuned the **UR5 arm** using TF transforms and PID control, validating autonomous navigation and pick-and-place workflows simultaneously in **Gazebo** and **RViz**.

MINI PROJECTS

- **Radix-2 8-Point Pipelined FFT | Verilog |** Github**:**
- Designed a **3-stage pipelined Radix-2 DIF FFT architecture** with modular butterfly, complex arithmetic, and twiddle factor units.
- Implemented **fixed-point streaming dataflow** using bit-reversed input ordering, double-buffered input storage, and valid handshaking for continuous FFT computation.

- **Pipelined FIR Filter Implementation | Verilog |** Github**:**
- Designed and implemented a parameterized **FIR filter in Verilog** with multiple architectures including direct form, optimized symmetric computation, and a pipelined systolic structure to study area-latency trade-offs.
- Developed an optimized architecture leveraging **coefficient symmetry and parallel accumulation**, reducing the number of multiplications and improving hardware efficiency compared to the baseline implementation.
- Built a **streaming input FIR filter** using a **systolic chain** tap units, enabling continuous data processing with fully pipelined throughput suitable for real-time DSP applications.

- **Eigenvalue Solver for Complex Matrices | C |** Github**:**
- Householder Hessenberg reduction followed by shifted QR iteration using Givens rotations for eigenvalue computation of complex matrices
- Complex-valued matrix support with stable convergence detection and upper-triangular thresholding for termination
- Eigenvalue extraction from quasi-triangular form with special handling of 2×2 Jordan blocks via characteristic polynomial solution

- **Scientific Calculator using Numerical Methods | Arduino UNO |** Github**:**
- Designed a infix-to-postfix expression parser using **Shunting Yard algorithm** supporting operator precedence for scientific functions as well.
- Implemented **RK4** for trigonometric functions, **Euler's method** for exponentiation, and **Trapezoidal integration** for logarithms, eliminating reliance on standard math libraries.
- Built a **16×2 LCD-based calculator** with a **resistor-ladder keypad interface** to maximize input functionality on the Arduino Uno's limited GPIO pins.

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant, EE1080: Probability**
  Conducted practice sessions and designed and graded programming assignments.

- **Domain Head, Programming and ML Domain,** Robotix IITH

- **Core Member, Programming and ML Domain,** Robotix IITH